

Nonadiabatic transitions in a two-level quantum system: Pulse-shape dependence of the transition probability for a two-level atom driven by a pulsed radiation field, Phys. Rev. A **57**, 79 (1998).

- [17] N. Rosen and C. Zener, Double stern-gerlach experiment and related collision phenomena, Phys. Rev. **40**,

502 (1932).

- [18] See Supplemental Material at [URL will be inserted by publisher] for the experimental setup, coherence-limit and adiabatic-basis derivations, pulse-cutoff and pulse-duration analyses, and additional high-amplitude data.